Urban Risk System - Formula Usage Guide

1) Min-Max Normalization

Formula: $X_{norm} = (X - X_{min}) / (X_{max} - X_{min})$

Used In: Data Preprocessing Stage.

Purpose: Scale traffic, accident, weather, pollution data to 0-1 range before risk calculation.

2) Urban Risk Index (Weighted Sum)

Formula: $URI = w1*T + w2*A + w3*W + w4*P$ (where weights sum to 1)

Used In: Risk Engine Module.

Purpose: Combine normalized factors to compute overall zone risk score.

3) Mean (Monthly Average Risk)

Formula: $R_{bar} = (1/n) * sum(R_i)$

Used In: Seasonal Analysis Module.

Purpose: Calculate average monthly risk per zone.

4) Variance (Risk Stability Check)

Formula: $Var(R) = (1/n) * sum((R_i - R_{bar})^2)$

Used In: Seasonal Stability Analysis.

Purpose: Detect fluctuation intensity in risk over months.

5) Covariance

Formula: $Cov(X,Y) = (1/n) * sum((X - X_{bar})(Y - Y_{bar}))$

Used In: Trend & Correlation Analysis.

Purpose: Measure directional relationship between variables (e.g., rainfall and accidents).

6) Pearson Correlation

Formula: r = Cov(X,Y) / (sigma_X * sigma_Y)

Used In: Feature Relationship Study.

Purpose: Identify strength of relationship between two variables.

7) Linear Trend Slope

Formula: beta1 = Cov(X,Y) / Var(X)

Used In: Monthly Trend Detection.

Purpose: Detect increasing or decreasing risk pattern over time.

8) Linear Regression Model

Formula: y = beta0 + beta1*x1 + beta2*x2 + ... + betan*xn

Used In: Prediction Module.

Purpose: Predict next month's urban risk score.

9) RMSE

Formula: sqrt((1/n) * sum((y - y_hat)^2))

Used In: Model Evaluation.

Purpose: Measure prediction error magnitude.

10) MAE

Formula: (1/n) * sum(|y - y_hat|)

Used In: Model Evaluation.

Purpose: Measure average absolute prediction error.